



SpiralWave

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

SpiralWave Inc.

Company or organization location (we welcome applicants from anywhere in the world)

US company. The main lab is in Tirol, Austria. Business office in SF.

Name(s) of primary point(s) of contact for this application

Abed Bukhari

Brief company or organization description <20 words

SpiralWave is a visionary company, developing truly scalable, energy efficient and cost effective DACS and CCU solutions.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

SpiralWave reinvents the air-liquid interface in carbon capture systems with CaptureBeam, a DAC unit that integrates electrochemistry and physics in a unique way. While conventional liquid solvent approaches rely on spraying or pumping aqueous solutions through separate capture and regeneration units, CaptureBeam integrates both functions in a single, elegant interface. The

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

interface undergoes an electrochemical pH swing process that generates regions of high alkalinity and acidity capturing and releasing CO₂ continuously - without fluid pumping, batch processing or ion-exchange membranes.

CaptureBeam is composed of cost effective material, inputs electricity, water, and stimulated high airflow, and operates at ambient conditions. The interface is air-permeable which enables three notable benefits: airflow, pumped using a fan, moves freely with no pressure drop, ideal gas-liquid contact - without solvent pooling or residence time limitations experienced in packed beds and spray tower contactors - and access to ambient oxygen which allows for selective oxygen reduction reaction at the cathode, avoiding unwanted hydrogen evolution. Moreover, the electrochemical process maintains a low current density, sidestepping heat generation, side reactions, and expensive noble metal electrodes.

Once released, the CO₂ is routed for compression and permanent storage via mineralization or geological injection pathways. These storage methods ensure atmospheric removal lifetimes exceeding 1,000 years, with well-established permitting, tracking, and MRV infrastructure.

Our system's advantages, which make us the best in class, manifest in three core parameters in DAC: energy efficiency, material cost, and scalability. Moreover, we identify a fourth core parameter, unique to electrochemical systems, intermittent renewable energy compatibility.

1- Energy Efficiency: CaptureBeam eliminates energy-intensive processes that constrain conventional systems, including:

- No mechanical pumping of liquid solvents.
- No pressure drop across the interface, reducing fan energy consumption.
- Selective oxygen reduction through an open, air-transparent configuration.
- Low current density operation that minimizes heat generation, side reactions and extends component lifetime.
- 100% solvent utilization rate.
- All-electric operation compatible with variable renewable energy sources

2- Material Cost: Our fundamentally simplified platform enables significant capital cost reductions:

- Integrated CO₂ capture and regeneration in a single interface eliminates separate absorber/desorber units.
- Unlike traditional electrochemical systems, our technology does not require AEM, PEM, or other membranes, nor noble metal electrodes.
- Uses widely available, non-precious electrode materials.

3- Innovative Scalability: CaptureBeam's unique design enables a fundamentally different approach to scaling:

- Processing capacity increases 3-dimensionally without increasing the electrode's surface area.
- Compact physical footprint supports diverse deployment scenarios
- Adaptable to various operational requirements and site-specific conditions

4- Renewable Energy Compatibility: Our system is inherently aligned with renewable electricity markets and operates without bulky solvent storage, or mechanical fluid handling systems.

Differentiation from Competitors

What truly sets SpiralWave apart from all other organizations working in the DAC space is truly our superior technology foundation. Today, DAC is riddled with multi-step workflows, thermal and pressure-based processes, significant pressure drop in airflow, linear scaling costs, expensive materials—ranging from membranes to rare earth metals and specialty chemicals—and a host of inefficiencies and complexities all-around.

SpiralWave's approach, enabled by the novel, low-cost air-liquid interface, allows seamless ambient condition capture and regeneration across a large surface area with no pressure drop, high current

densities or solvent underutilization. Perhaps most notably, our 3D scale up approach allows us to benefit from economies of scale, allowing for the increase of capacity without the proportional increase of hardware cost. This unique integration of physics and electrochemistry is a paradigm shift. Moreover, CaptureBeam is not constrained by a limited or difficult to access supply of feedstock. As of today, It is truly the only practical pathway towards rapid, cost-effective, and distributed, gigaton-scale climate impact.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

SpiralWave is commencing a commercial pilot project featuring an expansion of our CaptureBeam prototype into a 20-foot container capable of removing 200 tons of CO₂ annually. The CaptureBeam container is designed for intermittent renewable energy integration, and continuous and autonomous operation with minimal infrastructure requirements. The initial deployment will be outside our facility located in **Wörgl, Austria**, where we operate a 5,400 ft² lab that includes in-house mechanical, electrical, and chemical analysis capabilities. This controlled environment allows us to iteratively test and refine critical system components to achieve a commercial-grade design optimized for cost, scalability, and versatile field deployment. Upon achieving a commercial-ready unit, we will deploy our container in collaboration with a third-party mineralization provider.

To achieve Frontier's annual 0.5 Gt CO₂ removal target, we plan to mass manufacture and both run & sell the containers to DACS developers. We envision the purchase of our DAC containers to be as simple as buying a megascale battery from Tesla.

Cost Breakdown and Path to Cost Reduction

>Current Cost Breakdown:

- CapEx:

Capital expenses include the containerized DAC unit, and integrated control systems. While the system is designed for modularity and low material intensity, early-stage deployments incur higher costs due to small-batch production and custom integration. CapEx per ton for this project is \$631/tCO₂, dropping significantly to \$16/tCO₂ at NOAK with 50K ton capacity.

- OpEx:

The largest share of current costs arises from operational inputs, particularly electricity for electrochemical operation and air handling. Additional OpEx contributors include labor, system monitoring, maintenance and improvements. These costs reflect pilot-scale inefficiencies that are expected to decline significantly with scale and automation. Our Opex for this project is \$134/tCO₂, decreasing significantly to \$30/tCO₂ at NOAK scale.

- Sequestration and Storage:

SpiralWave partners with sub-surface based mineralization to ensure permanent storage of captured CO₂. This includes co-location, liquefaction, sequestration, and MRV, contributing to the total cost of carbon removal.

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

>Realizing Frontier's <\$100 cost target through the following Cost Reductions:

- **Manufacturing Scale-Up:**

Automating assembly and procuring core components at higher volumes will reduce per-unit costs

- **Energy Efficiency & Load Matching:**

Electrochemical optimization, paired with dynamic operation during periods of low-cost renewable electricity, will lower energy use per ton of CO₂ captured.

- **Automation & System Integration:**

Transitioning from pilot-phase manual oversight to fully automated operation will reduce labor, improve uptime, and enable remote performance monitoring at scale.

- **Sequestration Co-Location:**

Deploying capture units adjacent to mineralization or industrial storage sites will reduce transport and compression requirements, streamlining post-capture logistics.

- **Scaling up CaptureBeam and sequestration will result in economies of scale.**

CaptureBeam units are designed to scale vertically, enabling compounding cost reductions as the same electronic setup can handle multiple production capacities. Moreover, through larger volume sequestration, compounding cost reductions will be realized.

Quantifying Carbon Removed (MRV Approach)

SpiralWave uses a mass-balance-based MRV approach that combines real-time process monitoring with third-party-verified storage validation. CO₂ capture is quantified through continuous measurement of inlet and outlet concentrations using high-precision NDIR sensors, paired with calibrated airflow and mass flow metering. Released CO₂ is metered prior to compression and transfer to mineralization partners, who provide verifiable documentation of storage fate, including material type, location, and permanence. All data is timestamped, auditable, and aligned with ISO 14064 and emerging CDR industry best practices to ensure transparency, traceability, and durability of carbon removal claims.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

We perceive the following risks for our Initial Pilot project, categorized by key dimensions:

Technical:

(1) Electrode lifetime and interface stability under continuous operation are important to long-term system performance and cost. While initial lab tests show favorable durability, our ingenious team identifies and tests a suite of material and applies proprietary techniques to ensure longevity.

(2) Maintaining stable solvent circulation and utilization across varying environmental conditions presents a challenge in open-air operation. Our mitigation includes real-time monitoring and control of electrochemical inputs and environmental variables during pilot deployment to fine-tune feedback systems.

Project Execution:

(1) Our Pilot project will be derisked through incremental development stages. It is initially deployed within our own lab in Wörgl, Austria, minimizing dependency on third-party infrastructure during the critical system integration phase. This step will bring our technology to TRL 6 and be a further derisking point for prospect offtake and sequestration partners.

(2) The carbon sequestration market is dominated by large corporations, placing a high barrier for startups to secure partnerships. We mitigate this risk by collaborating with startups that validated their technology.

Measurement, Reporting, and Verification (MRV):

CO₂ storage is handled by mineralization partners. We mitigate custody-chain risks by requiring third-party documentation of storage location, material type, and carbon permanence, aligned with Frontier’s best practices.

Ecosystem:

CaptureBeam operates at ambient conditions with no chemical emissions, water discharge, or ecosystem disturbance. The system uses only air, water and electricity as inputs and is constructed from inert, non-toxic materials. As a result, we anticipate negligible negative ecological or agricultural impacts in any environment.

Financial:

First of a kind projects are inherently expensive due to their novel nature and the unforeseen challenges that often arise. To mitigate these substantial costs, we have adopted a two-pronged approach: maintaining a lean, highly-skilled team, and conducting the initial phase of the pilot within the controlled environment of our fully-equipped laboratory.

By keeping our team small and focused, we minimize overhead costs and streamline decision-making processes, ensuring efficient use of resources. Additionally, our state-of-the-art laboratory provides a cost-effective and controlled setting for the pilot project, eliminating the need for expensive external facilities and allowing us to rapidly identify and address any technical issues that may arise.

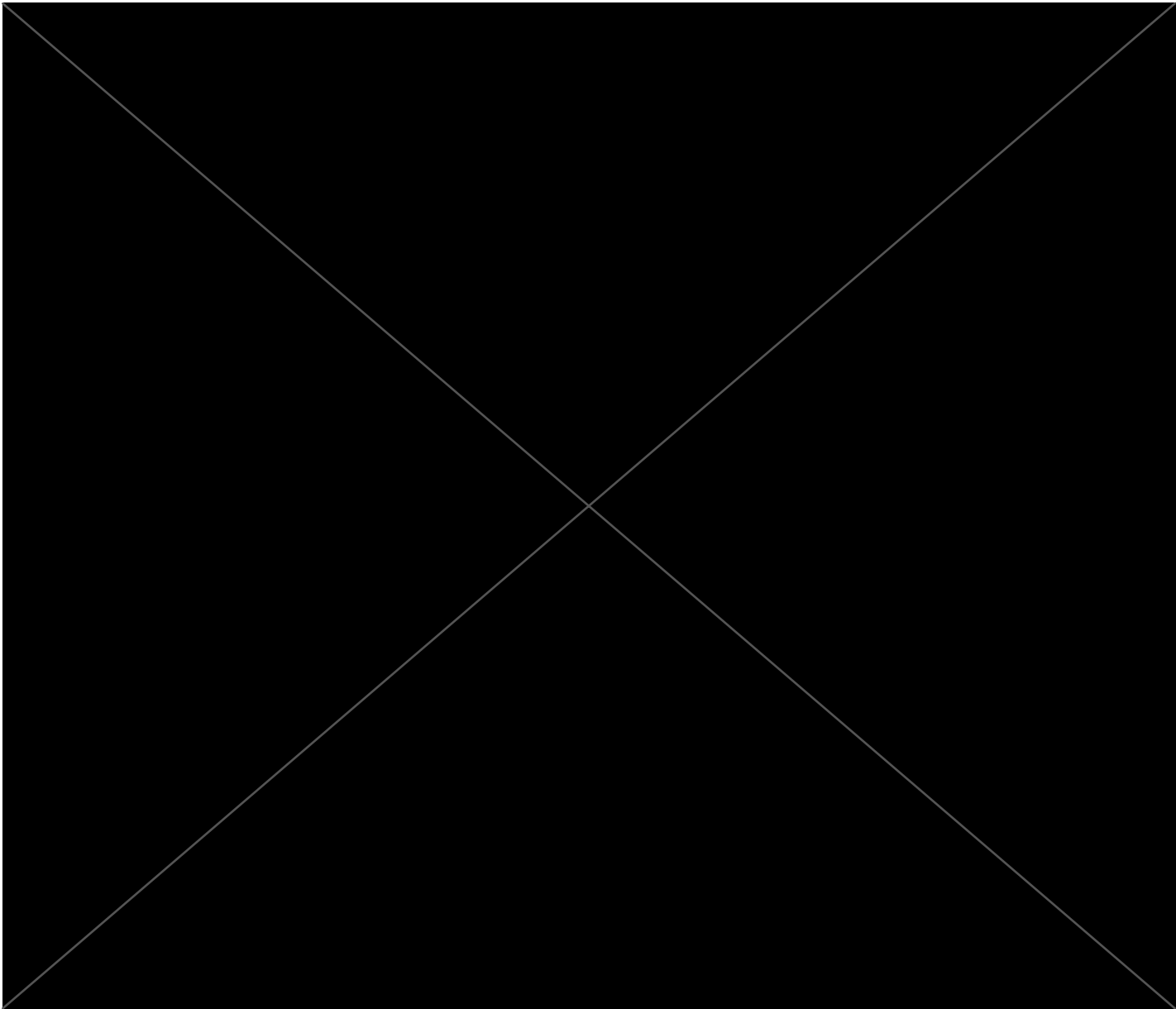
Other / Market Risk:

There is general uncertainty around voluntary demand and long-term pricing for high-quality CDR. SpiralWave is mitigating this risk by focusing on a low-complexity, modular platform that can scale incrementally and serve distributed markets. Moreover, we developed a carbon utilization solution, converting the captured carbon into methanol, to ensure larger demand.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	500
Delivery window	Q4 2026 - Q4 2028

(at what point should Frontier consider your contract complete? Should match 2f)	
Levelized cost (\$/ton CO ₂) (This is the cost per ton for the project tonnage described above, and should match 6d)	875
Levelized price (\$/ton CO ₂) ³ (This is the price per ton of your offer to us for the tonnage described above)	1000



³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).